

The degree of regularity of Weil-descent systems for the third summation polynomial over $\mathbb{F}_{2^n}$: exact closure dimensions, thresholds and a growth law

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Abstract

Huang, Kusters and Yeo (CRYPTO 2015) asked how the degree of regularity of the Weil descent of the third summation polynomial $S_3(X_0, X_1, x(Q))$ over $\mathbb{F}_{2^n}$, with X_0, X_1 restricted to a random $\mathbb{F}_2$ -subspace of dimension $\lceil n/2 \rceil$, grows with n ; their Magma experiments reached $n = 30$ (degree 4) and $n = 40$ (degree ≥ 5). We revisit the question with three independent tools: an exact computation of the degree-capped closures V_D of their Definition 1 (a C implementation reaching 32 variables at degree 5), a corrected build of the M4GB Gröbner-basis solver, and a Buchberger reference implementation. We find that (i) the degree of regularity is a deterministic function of the pair (N, n) of variables and equations, independent of the curve, the subspace, the point and the number of solutions, and that the $\lceil n/2 \rceil$ convention shifts the thresholds down for odd n ; (ii) the dimensions of the generic closures are exact polynomials, $\dim V_3 = (3N-2)n - \binom{N}{2}$, $\dim V_4 = \frac{1}{2}[3N(N+1)n - 171n - 9n^2]$, and the degree-5 part $Q_5 = 3n\binom{N}{3} - \binom{3n+1}{2}N - n(79N - 660)$, verified to the unit on 9 generic and 6 collapsed systems with $N \leq 32$; (iii) degree D suffices exactly when the generic degree- D part exceeds $\binom{N}{D}$, a criterion validated on 59 boundary cases and 110 grid outcomes without exception, including margins of 0.09% and 0.15%; (iv) the ideal contains exactly $3n$ quadrics — the n equations, $N - 1$ multiples of the trace-linear form and $2n - N + 1$ “mutants” — and the closures behave like a semi-regular system of $3n$ Boolean quadrics minus a small lag. Consequently the degree of regularity is 3 for $n \leq 16$, 4 for $17 \leq n \leq 36$, 5 for $37 \leq n \leq 52$ (the last value predicted, the value 5 at $n = 40$ confirmed) and grows linearly with slope $\approx 1/20$ in this range and 0.036 asymptotically, refuting the constant-degree first-fall-degree assumption for this family while confirming that the two-point decomposition offers no subexponential attack. We also report and fix a deviation from the Gebauer–Möller criteria in M4GB that inflates the reported step degree.

Notation. N is the number of Boolean variables and n the number of quadratic equations (the extension degree); $\mathcal{B} = \mathbb{F}_2[x_1, \dots, x_N]/(x_i^2 + x_i)$ is the Boolean polynomial ring, $\mathcal{B}_{\leq k}$ its subspace of polynomials of degree $\leq k$, $|\mathcal{B}_{\leq k}| = \sum_{j \leq k} \binom{N}{j}$; $\binom{N}{\leq k}$ abbreviates the same sum.

1 Introduction

Index calculus on elliptic curves over binary fields $E(\mathbb{F}_{2^n})$ rests on the *point decomposition problem*: given $Q \in E(\mathbb{F}_{2^n})$, find P_0, P_1 in a factor base $\mathcal{F}_V = \{P : x(P) \in V\}$, V an $\mathbb{F}_2$ -subspace of $\mathbb{F}_{2^n}$, with $P_0 + P_1 = Q$. Semaev’s third summation polynomial $S_3(X_0, X_1, x(Q))$ turns this into a bivariate equation over $\mathbb{F}_{2^n}$; writing X_0, X_1 in a basis of V and taking coordinates over $\mathbb{F}_2$ (Weil descent) turns it into a system of n quadratic equations in $N = 2 \dim V$ Boolean variables [24, 14, 7, 23]. The cost of the whole attack is governed by the *degree of regularity*

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D_{reg} of these systems — the maximal degree that a Gröbner-basis algorithm must reach — and the central open question, posed explicitly by Huang, Kusters and Yeo [17, Sec. 5.2], is how D_{reg} grows with n . The *first fall degree assumption* of Petit and Quisquater, under which D_{reg} stays close to the first fall degree (here 2 or 3) for all n , would make the attack subexponential; HKY’s experiments in Magma ($n \leq 40$) showed D_{reg} increasing from 3 to at least 5 without settling the rate, and Kusters and Yeo [20] added $n = 17$ and $n = 30$ and reported that $n = 40$ exhausted 38 GB of memory before finishing.

This note takes the question as far as computation allows, and then one step further by a structural model. Our contributions are the following.

1. **Deterministic dependence on (N, n) .** For random subspaces the degree of regularity depends only on the number of variables N and equations n : it is independent of the curve, the subspace, the point Q , the number of solutions (including inconsistent systems), and of the choice of instance. Under the convention $\dim V = \lceil n/2 \rceil$ of HKY, odd n has one variable more than equations ($N = n + 1$) and reaches the next degree earlier than even n — the source of the apparently irregular entries $15 \rightarrow 4$, $35 \rightarrow 5$ in our tables.
2. **Exact closure dimensions.** Let V_D be the degree-capped closure of the system (HKY, Definition 1). In the *generic* regime — before the closure collapses onto the whole truncated ideal — its dimensions are exact polynomials in (N, n) :

$$\dim V_3 = (3N - 2)n - \binom{N}{2}, \quad \dim V_4 = \frac{1}{2}[3N(N + 1)n - 171n - 9n^2],$$

and the degree-5 part of V_5 is $Q_5 = 3n\binom{N}{3} - \binom{3n+1}{2}N - n(79N - 660)$. These were fitted on a handful of points and then verified to the unit on 9 further generic points at $N \leq 32$ ($\dim V_5$ up to 221 331) and on 6 collapsed points, with margins down to 0.15%.

3. **A threshold criterion.** Degree D suffices — the reduced Gröbner basis lies in V_D , so no F4-type algorithm needs a higher step degree — exactly when the generic degree- D part exceeds the number of degree- D monomials: $Q_D(N, n) > \binom{N}{D}$. The criterion was tested on 59 boundary cases and 110 further grid outcomes (square, odd and unbalanced systems; closure computations and two Gröbner-basis solvers) without exception. It gives $D_{\text{reg}} = 3$ for $n \leq 16$, 4 for $17 \leq n \leq 36$ and 5 for $37 \leq n \leq 52$, the last value being a prediction; direct computations reach $n = 40$ (M4GB, 6 h) and confirm the value 5 that Magma could not reach [20].
4. **Structure and a growth law.** The ideal contains exactly $3n$ quadratic polynomials: the n equations, $N - 1$ multiples of the trace-linear form of HKY’s Proposition 5, and $2n - N + 1$ further quadrics arising as degree falls. The closures behave, up to an explicit lag, like the Macaulay matrices of a *semi-regular* system of $3n$ Boolean quadrics in N variables [2, 3]: the semi-regular thresholds are 20, 37, 56, 76, 97, 119, ... for $D = 3, 4, 5, \dots$, ours are 16, 36, 52. Consequently D_{reg} grows linearly in n , with about one degree per 20 variables in the computed range and slope 0.036 asymptotically — neither constant, as the first fall degree assumption would have it, nor logarithmic. The multiplier 3 explains why these systems are much easier than generic systems of n quadrics (slope 0.09) while still exponentially hard.
5. **A solver defect.** The reported step degree of the Gröbner-basis solver M4GB [21] is inflated by one on part of this family because its pair-update discards both members of every pair of critical pairs with equal lcm, whereas the Gebauer–Möller criteria keep one. We locate, reproduce and fix the defect (pull request #14); the corrected solver agrees with the closure computations everywhere.

We stress what is *not* claimed. The formulas are empirical: exact on every point we could compute, with the structural explanation of Section 7 reaching the first of the two stages of the closure. Beyond $D = 5$ the thresholds are predictions of the semi-regular model with a lag that we can bound but not derive. No cryptographic consequence follows for the decomposition with two points: even a bounded D_{reg} would leave the index calculus at $2^{n/2}$ relations for a factor base of this size.

2 Preliminaries

2.1 Curves, summation polynomial, Weil descent

Let $k = \mathbb{F}_{2^n}$ and let $E : y^2 + xy = x^3 + a_2x^2 + a_6$ be an ordinary elliptic curve over k (every ordinary curve over k is isomorphic to one of this form; the isomorphism acts on x by an affine map and does not affect the degrees considered below). The third summation polynomial of E in characteristic 2 is, from the general formula of [17, Sec. 5.1] with $b_2 = 1$, $b_4 = b_6 = 0$, $b_8 = a_6$,

$$S_3(X_0, X_1, X_2) = X_0^2X_1^2 + X_0^2X_2^2 + X_1^2X_2^2 + X_0X_1X_2 + a_6,$$

with the property that $S_3(x(P_0), x(P_1), x(P_2)) = 0$ if and only if $P_0 \pm P_1 \pm P_2 = O$ for some choice of signs; we verified this on random triples for every field used.

Fix a subspace $V \subset k$ of dimension $d = \lceil n/2 \rceil$ with basis $v_1, \dots, v_d$ and a point $Q \in E(k)$. Substituting $X_0 = \sum u_i v_i$, $X_1 = \sum w_j v_j$ with $u_i, w_j \in \mathbb{F}_2$ and expanding $S_3(X_0, X_1, x(Q))$ in a basis of k over $\mathbb{F}_2$ gives n polynomials in $N = 2d$ variables. Because squaring is $\mathbb{F}_2$ -linear, $X_0^2X_1^2$ and $X_0^2x(Q)^2$ contribute only bilinear and linear monomials: the system is

$$f_r(u, w) = \sum_{i,j} c_{r,ij} u_i w_j + \sum_i d_{r,i} u_i + \sum_j e_{r,j} w_j + a_{6,r}, \quad r = 1, \dots, n,$$

with $c_{ij} = v_i^2 v_j^2 + v_i v_j x(Q)$, $d_i = v_i^2 x(Q)^2$, $e_j = v_j^2 x(Q)^2$ (coordinates over $\mathbb{F}_2$). Together with the field equations $u_i^2 = u_i$, $w_j^2 = w_j$ this is the system studied in [17, Sec. 5.2]; we work in the Boolean ring $\mathcal{B}$, where the field equations are built in. The solution set is the set of decompositions of Q ; by the symmetry $X_0 \leftrightarrow X_1$ it has even size unless $X_0 = X_1$ is possible. For even n the system is square ($N = n$); for odd n it has one variable more than equations ($N = n + 1$).

HKY take Q to be a random point, so that about 60% of their systems have no solution; we generate both random- Q systems and *consistent* systems $Q = P_0 + P_1$ with P_0, P_1 drawn uniformly from $\mathcal{F}_V$, and find no difference in any degree measured. We also consider unbalanced systems in which V has dimension $d > \lceil n/2 \rceil$ or the two points use subspaces of different dimensions $d_0 + d_1 = N$; the family is parametrised by (N, n) with $n \geq \max(d_0, d_1)$, hence $N \leq 2n$.

2.2 Degree-capped closures and degrees

For a set F of polynomials in $\mathcal{B}$ and $D \geq 2$, the closure $V_D = V_{F,D}$ is the smallest $\mathbb{F}_2$ -subspace containing every $f \in F$ of degree $\leq D$ and closed under $g \mapsto x_a g$ whenever $\deg g \leq D - 1$ ([17, Definition 1], transported to the Boolean ring; the two definitions agree modulo the field equations). We call the *first fall degree* the least c with $V_c \cap \mathcal{B}_{\leq c-1} \neq V_{c-1}$, the *last fall degree* the greatest such c , and the *solving degree* D_{solve} the least D such that V_D contains the reduced Gröbner basis of the ideal (for a degree-compatible order; cf. [6, 5]). For the systems at hand the first fall degree is always 2 (Proposition 5 of [17]: a fixed linear combination of the f_r is the linear polynomial $\text{Tr}((x(Q) + a_2)/a_1^2) + \text{Tr}(X_0 + X_1) = 0$), and the last fall degree coincided with D_{solve} in every instance we computed.

HKY define the degree of regularity operationally as the largest step degree of Magma’s F4 at which new polynomials are produced. For any F4-type algorithm with the normal selection strategy, every polynomial produced at step degree $\leq D$ lies in V_D ; hence

Lemma 1. $D_{\text{solve}} \leq D_{\text{reg}}$ for every such algorithm.

The inequality is an equality for Magma’s F4 on all of HKY’s data points and for the corrected M4GB on all of ours; the uncorrected M4GB violates it by one on part of the family (Section 3.3). We therefore use D_{solve} — a property of the system, not of a program — as the primary quantity and report the solver’s degree as a check.

2.3 Testing containment of the Gröbner basis

Let E_D be an echelon basis of V_D with respect to a monomial order refining degree (we use grevlex) and let $\text{LM}(V_D)$ be its set of leading monomials. Write $\text{std}(S)$ for the number of squarefree monomials divisible by no element of a set S .

Lemma 2. V_D contains the reduced Gröbner basis of the ideal J generated by F if and only if $\text{std}(\text{LM}(V_D))$ equals the number of solutions of F in $\mathbb{F}_2^N$.

Proof. J is radical and $\mathcal{B}/J$ has dimension $\#V(F)$, so $\text{std}(\text{LM}(J)) = \#V(F)$; $\text{LM}(V_D) \subseteq \text{LM}(J)$ gives $\text{std}(\text{LM}(V_D)) \geq \#V(F)$, with equality iff the two monomial ideals coincide. If they coincide, every minimal generator m of $\text{LM}(J)$ is a leading monomial of V_D ; every non-standard monomial of degree $\leq D$ is then also in $\text{LM}(V_D)$ (multiply the element of V_D whose leading monomial divides it by the complementary squarefree variables — the degree bound is respected and, in the Boolean ring, the product’s leading monomial is the intended one), so the fully reduced row of E_D with pivot m has only standard monomials in its tail and equals the reduced Gröbner-basis element with that leading monomial. $\square$

The number of solutions is obtained exactly by exploiting the bilinear structure: for each of the 2^{d_0} values of u the system is affine in w and is solved by linear algebra.

3 Tools and methodology

3.1 Instance generation

Field arithmetic over $\mathbb{F}_{2^n}$ (irreducible polynomials found by the Ben-Or test), curve arithmetic and the derivation of S_3 were implemented from scratch and checked against the group law on random points; the Weil descent was checked by evaluating every generated system on its known solution. Subspaces are random (a uniformly chosen independent set); coordinates of points in V are computed by Gaussian elimination. All instances are reproducible from a seed; the seeds of every instance in this note are listed in the supplementary archive.

3.2 Closure computations

Closures were computed by two independent programs. The first, in Python, represents Boolean polynomials as bit vectors over the monomials of degree $\leq D$ in grevlex order and computes V_D by iterated multiplication with echelon reduction; it handles $N \leq 26$ at $D = 4$ and was used for all $D = 3, 4$ data and for the verification of Lemma 2 by comparison with the Buchberger–Möller basis of the vanishing ideal of the solution set. The second, in C (about 400 lines), stores rows as packed 64-bit words up to their pivot, keeps the echelon block-triangular (every pivot row has zeros at the pivot columns of earlier blocks; the rows of one block are in reduced echelon form among themselves), and reduces batches of new rows by Gray-code tables of $k = 10$ pivot rows (the Method of Four Russians), so that no reduced-row-echelon maintenance

is needed when pivots are appended. Rows are expanded once, in insertion order, and only if their degree is $\leq D - 1$; with OpenMP the reduction of a batch is parallel over rows. It reproduces every Python result to the unit and reaches $N = 32$ at $D = 5$ (243 000 columns, 3.7 GB, 5–13 hours on eight threads). A second mode computes the rank of the plain product set $\{tq : q \in W_2, \deg t \leq D - 2\}$ of Section 7, without the iterated expansion of degree falls.

3.3 Gröbner-basis solvers

We used M4GB [21] (GF(2), field equations included as input polynomials; Cygwin, GCC 13, -03) as the Gröbner-basis solver, reading D_{reg} from its per-step output as the largest lcm degree of a step reporting new polynomials — the definition of HKY. Two remarks are necessary for reproducibility. First, the libtool build places a 27KB wrapper in `bin/` and the actual executable in `bin/.libs/`; renaming the wrapper does not preserve a build, and the executable in `.libs/` must be copied after each build with a different `MAXVARS`. Second, in the pair update of `m4gb.hpp` the Gebauer–Möller M-criterion is implemented with non-strict divisibility of lcms,

```
if (it != it2 && it2->first.intlcm | it->first.intlcm)
    { it->second = false; break; }
```

so that two admissible new pairs with equal lcm discard each other, whereas the F-criterion keeps exactly one (none if a coprime pair shares the lcm) [15]. The final basis was correct in every run we made — the lost S-polynomials reappear through pairs of higher lcm degree — but the maximal step degree is inflated by one where those S-polynomials carry the degree falls. On the $n = 16$ systems of HKY the solver reports 4 (13 new polynomials at degree 4: exactly the 13 linear polynomials that complete the basis) against HKY’s 3; with the criterion corrected (a 20-line change; the fix is available as pull request #14 to the M4GB repository, <https://github.com/cr-marcstevens/m4gb/pull/14>, and as a standalone patch in the supplementary archive) it reports 3 on all 41 instances we tested, with the same basis, and agrees with D_{solve} on every instance of this note. We reproduced the mechanism independently with a Buchberger implementation whose pair criteria can be switched between the two versions.

3.4 Verification chain

Each number in Sections 4–6 rests on at least two of: the closure computation (D_{solve} by Lemma 2), the corrected M4GB (D_{reg} by HKY’s definition), the Buchberger reference implementation, and, for HKY’s own sizes, their published Magma values. Where a polynomial formula is stated, the points used to fit it are separated from the points used to verify it, and the latter — 21 in total — were computed after the prediction was recorded. The threshold predictions of Section 6 were likewise written down, with their margins, before the corresponding systems were run.

4 Results for the systems of HKY

Table 1 collects the degree of regularity for the systems of [17, Sec. 5.2] — random curve, random subspace of dimension $\lceil n/2 \rceil$ — from $n = 12$ to $n = 40$. “Closure” is D_{solve} computed by Lemma 2; “M4GB” is the corrected solver’s largest productive step degree; the last two columns are the published values of HKY (Magma F4) and KY. For $n \leq 24$ the closure column summarises 378 instances (random and consistent Q , fixed and varying curves and subspaces, several instances per n) with no variation whatsoever; for $n \geq 26$ two instances were run at 26–32, 35 and 36 and one otherwise. Entries in parentheses are the values given by the criterion of Section 6; the closures at these sizes were not computed directly (the degree-4 closure was computed up to $N = 30$ for unbalanced systems, Table 4).

Table 1: Degree of regularity of the Weil-descent system of S_3 , $\dim V = \lceil n/2 \rceil$. Bold: odd n at which the degree increases one step earlier than for the neighbouring even n .

n	N	closure	M4GB (fixed)	HKY	KY	time / memory (M4GB)
12	12	3	3	3	3	< 1 s
13	14	3	—			
14	14	3	3			< 1 s
15	16	4	—			
16	16	3	3 (4 before the fix)	3	3	< 1 s
17	18	4	—		4	
18	18	4	4	4		1 s
19	20	4	—			
20	20	4	4	4	4	3 s
22	22	4	—			
24	24	4	—	4		
26	26	4	4			26 s / 7 MiB
28	28	(4)	4			71 s / 7 MiB
30	30	(4)	4		4	214 s / 7 MiB
32	32	(4)	4			8 min / 9 MiB
33	34	(4)	4			18 min / 0.6 GiB
34	34	(4)	4			17 min / 0.6 GiB
35	36	(5)	5			36–44 min / 0.6–1.0 GiB
36	36	(4)	4			38–39 min / 0.5–0.8 GiB
37	38	(5)	5			82 min / 1.0 GiB
38	38	(5)	5			78 min / 0.8 GiB
39	40	(5)	5			5.1 h / 2.2 GiB
40	40	(5)	5		≥ 5	6.0 h / 1.5 GiB

Three observations organise the rest of the paper. (i) The degree is a function of (N, n) only: 378 instances at $n \leq 24$ — solvable and unsolvable, with 0 to 8 solutions, with the same or different curves and subspaces — give identical closure dimensions in every degree, not merely the same D_{solve} (Section 5). (ii) With $N = n + 1$ for odd n , the thresholds are crossed one step earlier: $15 \rightarrow 4$ while $16 \rightarrow 3$, and $35 \rightarrow 5$ while $36 \rightarrow 4$; the natural coordinates are (N, n) , and in them there is no irregularity. (iii) The values 3, 4, 5 are reached at $n = 12$ –16, 17–36, 37–40 respectively, consistent with HKY and KY at every common size and extending KY’s “ ≥ 5 ” at $n = 40$ to an exact 5.

The unsolvable systems of HKY’s construction (random Q) deserve a remark: the closure then contains 1, and the criterion of Section 6 applies verbatim — the “degree of regularity” of an inconsistent system is the degree at which 1 is reached, and it coincides with the degree of the consistent systems of the same (N, n) in all 60 cases we compared.

5 Generic closure dimensions

5.1 Profiles

For every instance we computed the full profile $\dim(V_D \cap \mathcal{B}_{\leq k})$, $k \leq D$. In the *generic* regime — as long as V_D is a proper subspace of the truncated ideal $J_{\leq D}$ — the profiles are the same for all instances with the same (N, n) , and they are polynomials in (N, n) . In the *collapsed* regime $V_D = J_{\leq D}$, of dimension $|\mathcal{B}_{\leq D}|$ minus the number of solutions (all standard monomials of these ideals have degree ≤ 2), and $D_{\text{solve}} \leq D$.

The lowest levels are uniform across the family:

- $\dim V_2 = n + N - 1$: the n equations plus the $N - 1$ independent multiples $x_a \ell$ of the trace-linear polynomial $\ell \in \text{span}(f_r)$.

- $\dim(V_3 \cap \mathcal{B}_{\leq 2}) = 3n$: the space W_2 of *all* quadratic polynomials of the ideal (it equals $J_{\leq 2}$, as confirmed by Buchberger–Möller on the solution sets) has dimension exactly $3n$. It consists of the n equations, the $N - 1$ multiples of ℓ , and $M_3 = 2n - N + 1$ further quadrics obtained as degree falls from cubic products. No further quadric appears at any higher level.
- $\dim(V_4 \cap \mathcal{B}_{\leq 3}) = 3n(N + 1)$, and $\dim(V_5 \cap \mathcal{B}_{\leq 4}) = \dim V_4 + 8n$.

5.2 The polynomials

Table 2: Generic closure dimensions. $P_D = \dim V_D$; $Q_D = \dim V_D - \dim(V_D \cap \mathcal{B}_{\leq D-1})$ is the degree- D part.

level	formula	fitted on	verified on
P_3	$(3N - 2)n - \binom{N}{2}$	$N = 16, 18$ (slope and intercept in n)	40 further (N, n) , $N = 14$ –26, exact
P_4	$\frac{1}{2}[3N(N + 1)n - 171n - 9n^2]$	$N = 18, 22, 24$	$N = 20, 26, 28, 30, 32$ (13 points), exact
Q_5	$3n\binom{N}{3} - \binom{3n+1}{2}N - n(79N - 660)$	$(24, 12), (26, 13)$	7 points, $N = 28$ –32, $n - N/2 \leq 3$, exact

The data behind Table 2: for $D \leq 4$ a grid of 55 systems with $N = 16, 18, \dots, 24$ and n running from $N/2$ to N ($\dim V = N/2$ for both points, n equations from the field $\mathbb{F}_{2^n}$); a second family with $\dim V_0 = (n + 1)/2$, $\dim V_1 = (n - 1)/2$ ($N = n$ for odd n); and the square and odd systems of Table 1 up to $N = 26$. For $D = 5$, the 15 systems of Table 3.

Table 3: Level 5 (C program; dimensions to the unit). “Full” means $\dim V_5 = |\mathcal{B}_{\leq 5}| - \#\text{solutions}$. The first two rows fixed the two free parameters of Q_5 (the coefficients 79 and 660); every other entry was predicted before it was computed.

(N, n)	$\dim(V_5 \cap \mathcal{B}_{\leq 4})$	$P_4 + 8n$	$\dim V_5$	Q_5 measured	Q_5 formula	$\binom{N}{5}$
(24,12)	9 222	9 222	51 270	42 048	42 048	42 504
(24,13)	10 909	—	53 413 full	—	44 148	42 504
(26,13)	11 921	11 921	74 919	62 998	62 998	65 780
(26,14)	—	—	79 556 full	—	66 206	65 780
(26,15)	—	—	81 565 full	—	69 180	65 780
(28,14)	15 085	15 085	105 665	90 580	90 580	98 280
(28,15)	16 095	16 095	111 255	95 160	95 160	98 280
(28,16)	—	—	118 274 full	—	99 488	98 280
(30,15)	18 750	18 750	144 750	126 000	126 000	142 506
(30,16)	19 928	19 928	152 168	132 240	132 240	142 506
(30,17)	21 097	21 097	159 307	138 210	138 210	142 506
(30,18)	—	—	170 182 full	—	143 910	142 506
(32,16)	22 952	22 952	193 512	170 560	170 560	201 376
(32,19)	26 999	26 999	221 331	194 332	194 332	201 376
(32,20)	37 423	—	238 799 full	—	201 680	201 376

A second instance of (28, 15) and of (28, 16) gave the same numbers. The $8n$ new quartic polynomials at level 5 are visible in the second and third columns of Table 3.

5.3 What the closure is not

The closure is not the Macaulay matrix. The rank R_D of the plain product set $\{tq : q \in W_2, \deg t \leq D - 2\}$ is strictly smaller than P_D : at (20, 10), $R_4 = 2994$ against $P_4 = 4995$.

The difference is contributed by products of variables with *degree falls* — polynomials of degree $< D$ obtained as combinations of products whose top-degree parts cancel — which the iterated closure multiplies again. This two-stage structure is analysed in Section 7; it is the reason why P_D exceeds any single Macaulay-type count and why “semi-regularity” enters only after the falls are accounted for.

6 The threshold criterion

6.1 Statement

Observation 1 (criterion). *For $D = 3, 4, 5$ put*

$$\begin{aligned} Q_3(N, n) &= 3nN - 5n - \binom{N}{2}, \\ Q_4(N, n) &= 3n\binom{N}{2} - \binom{3n+1}{2} - 87n, \\ Q_5(N, n) &= 3n\binom{N}{3} - \binom{3n+1}{2}N - n(79N - 660). \end{aligned}$$

Then $D_{\text{solve}} \leq D$ if and only if $Q_D(N, n) > \binom{N}{D}$.

The “only if” direction is a tautology once the generic formulas are accepted: the degree- D part of V_D cannot exceed the number of degree- D monomials, so a generic value above $\binom{N}{D}$ is impossible and the closure must have left the generic regime, i.e. produced additional degree falls. The “if” direction — that *every* excess, however small, triggers a cascade that fills the entire truncated ideal — is an empirical fact: the smallest excesses observed to collapse are 8 out of 560 at (16, 16), 40 out of 46 376 at (34, 32) and 304 out of 201 376 at (32, 20), and no partial collapse was ever seen.

6.2 Tests

Table 4: Boundary cases (both sides of a threshold, or margin below 3%). c = collapse (degree D suffices), g = generic (degree $D + 1$ or more needed). “M4GB” cases were run with the corrected solver, “closure” cases by Lemma 2.

D	(N, n)	$Q_D - \binom{N}{D}$	predicted	observed	method
3	(14, 14), (15, 15), (16, 16)	+63, +40, +8	c	c	closure
3	(14, 13)	+26	c	c	closure
3	(16, 15), (17, 17), (18, 17), (18, 18), (20, 19)	−35, −34, −136, −87, −285	g	g	closure
4	(20, 11), (20, 12)	−93, +285	g, c	g, c	closure
4	(22, 13), (22, 14)	−217, +266	g, c	g, c	closure
4	(24, 15), (24, 16)	−546, +54	g, c	g, c	closure
4	(26, 16), (28, 16)	−1918, −4899	g	g	closure
4	(30, 24), (30, 25)	−801, +195	g, c	g, c	closure and M4GB
4	(28, 21), (28, 22)	−504, +348	g, c	g, c	M4GB
4	(32, 28), (32, 29)	−302, +841	g, c	g, c	M4GB
4	(34, 31), (34, 32)	−1271, +40	g, c	g, c	M4GB
4	(34, 33), (34, 34), (36, 36)	+1342, +2635, +117	c	c	M4GB
4	(36, 35), (38, 37), (38, 38), (40, 39), (40, 40)	−1365, −5217, −3534, −10426, −8530	g	g	M4GB
5	15 cases of Table 3	from −30 816 to +3 400		all as predicted	closure
5	(34, 22), (34, 23)	−3058, +5932	g, c	g, c	M4GB
5	(36, 24), (36, 25)	−9936, +1308	g, c	g, c	M4GB
5	(40, 30), (40, 31)	−7608, +8492	g, c	g, c	M4GB

Together with the 55 grid systems of Section 5 (each of which is a test at $D = 3$ and at $D = 4$) the criterion was checked on 59 boundary cases and 110 grid outcomes without a single failure; 21 of the boundary predictions were recorded, with their margins, before the systems were run. The M4GB tests at $D = 5$ are the most expensive computations of this note (up to 63 h and 8.5 GB at (40, 30)) and are also the ones that reach closest to the square systems: for (40, 30) and (40, 31) the ratio n/N is 0.75–0.78, against 0.50–0.59 for the points on which Q_5 was fitted.

6.3 Consequences for the family of HKY

For the square systems (even n , $N = n$) the criterion gives degree 3 up to $n = 16$, degree 4 for $18 \leq n \leq 36$ and degree 5 for $38 \leq n \leq 52$; for odd n ($N = n + 1$) degree 4 from $n = 15$, degree 5 from $n = 35$ and degree 6 from $n = 51$. The margins at the predicted transition are -0.24% at (52, 51), $+1.25\%$ at (52, 52), -3.1% at (54, 53) and -1.6% at (54, 54); the first of these is the only tight one. Every entry up to $n = 40$ is confirmed by computation; the values for $n = 41$ –54 are predictions of the polynomial Q_5 , extrapolated from $n/N \leq 0.78$ to $n/N = 1$. A direct test would require a Gröbner-basis computation with 52 variables reaching step degree 6, which is beyond M4GB on a workstation by two to three orders of magnitude; it is the natural target for a faster solver.

7 Structure of the closures and a growth law

7.1 Semi-regular part and lag

For m Boolean quadratic polynomials in N variables that are *semi-regular* in the sense of Bardet, Faugère, Salvy and Yang [2, 3], the dimension of the degree- k part of the Macaulay span is the coefficient

$$I_k^{\text{SR}}(N, m) = \binom{N}{k} - [z^k] \frac{(1+z)^N}{(1+z^2)^m} = m \binom{N}{k-2} - \binom{m+1}{2} \binom{N}{k-4} + \binom{m+2}{3} \binom{N}{k-6} - \cdots,$$

as long as this is positive; the first k for which the coefficient of the series becomes non-positive is the semi-regular degree of regularity. We verified this description on random systems: for 33 random dense Boolean quadrics in 22 variables the closure has the profile (33, 759, 7821) predicted by the series, with no correction.

Our polynomials Q_D are the semi-regular counts of a system of $m = 3n$ quadrics, minus a lag:

$$\begin{aligned} Q_3 &= I_3^{\text{SR}}(N, 3n) - \text{lag}_3, & \text{lag}_3 &= 5n + \binom{N}{2}, \\ Q_4 &= I_4^{\text{SR}}(N, 3n) - \text{lag}_4, & \text{lag}_4 &= 87n, \\ Q_5 &= I_5^{\text{SR}}(N, 3n) - \text{lag}_5, & \text{lag}_5 &= n(79N - 660). \end{aligned}$$

The identification of the Koszul-type term $\binom{3n+1}{2}$ in Q_4 and of $\binom{3n+1}{2}N$ in Q_5 with the second term of the semi-regular series is what makes this decomposition natural rather than a rewriting: the $3n$ quadrics are exactly the polynomials of W_2 (Section 5), and the closures behave as if these $3n$ polynomials were in generic position, up to a lag that is small relative to the leading term (26%, 5.0% and 6.4% of I_D^{SR} at the three thresholds). The lag is not a property of the ideal: the truncations $J_{\leq k}$ of the ideal itself are fixed by the solution set ($|\mathcal{B}_{\leq k}|$ minus the number of standard monomials of degree $\leq k$) and carry no generic information. It is a property of what the degree-capped closure can reach.

7.2 The first stage exactly

The closure has two stages: the span R_D of the plain products $\{tq : q \in W_2, \deg t \leq D-2\}$ and the products of variables with the degree falls that appear inside that span. The first stage admits an exact block decomposition. Write ℓ for the trace-linear polynomial, $m' = 3n - N$ for the number of quadrics of W_2 not in the ideal of ℓ , and $M_3 = 2n - N + 1$ for the number of degree falls at level 3 (“mutants” in the terminology of [10, 9]). Then

$$R_D = \binom{N-1}{\leq D-1} + \text{SR}_{\leq D}(N-1, m') - M_3 \binom{N}{\leq D-3} - \delta_D,$$

where $\binom{N-1}{\leq D-1} = \dim \ell \mathcal{B}_{\leq D-1}$ is the contribution of the ideal of ℓ , where $\text{SR}_{\leq D}(N-1, m') = \sum_{k=2}^D I_k^{\text{SR}}(N-1, m')$ is the semi-regular count of the remaining quadrics in the $N-1$ variables of the quotient by ℓ , the third term counts the defining relations of the mutants multiplied by monomials, and

$$\delta_3 = \delta_4 = 0, \quad \delta_5 = \binom{N}{2} - 1 - 3 \binom{M_3}{2} - \frac{(M_3 - 1)(N + 1)}{2}.$$

At $D = 4$ the formula reproduces the rank of the product set on all 7 systems where it was computed (2994 at (20, 10); 4025, 4636, 5238 at $N = 22$; 5269, 6008, 6738 at $N = 24$); at $D = 5$, with δ_5 fitted on (24, 12), (30, 15–17) and (32, 16), it reproduces R_5 on 9 systems, including the predicted values 46372, 63386 and 71178 at (26, 13), (28, 14), (28, 15) ($M_3 = 1, 1, 3$). At $D = 3$ the formula is an identity: $\binom{N-1}{2} + m'(N-1) - M_3 = 3nN - 5n - \binom{N}{2} = Q_3$, so Q_3 is *derived* from the block structure. The new term δ_5 has an evident reading — $\binom{N}{2} - 1$ relations between the ideal of ℓ and the other quadrics that first appear at degree 5, three Koszul-type relations per pair of mutants, and $(M_3 - 1)(N + 1)/2$ relations between mutants and ℓ — but we have not proved it.

The second stage, $S_D = P_D - R_D$, is known exactly as a difference of the formulas above ($S_4 = N^3/3 - N^2/2 + 2Nn - 5N/6 - 85n + 1$; S_5 of degree 4) and is the part of the closure that no Macaulay-type count captures: at (20, 10) it contributes 2001 of the 4365 degree-4 dimensions. Its structural derivation is open (Section 9).

7.3 Thresholds and the growth of D_{reg}

For square systems ($n = N$) the criterion $Q_D(N, N) > \binom{N}{D}$ gives the last N at which degree D suffices. Table 5 compares the semi-regular thresholds (lag = 0) with the actual ones.

Table 5: Last N with degree D for square systems.

D	semi-regular, $m = 3N$	with lag (this note)	ratio	lag/ I^{SR} at threshold
3	20	16	0.80	26%
4	37	36	0.97	5.0%
5	56	52 (predicted)	0.93	6.4%
6	76	66–76		
7	97			
8	119			

The semi-regular thresholds grow with gaps 17, 19, 20, 21, 22, 22, 23, ..., tending to the asymptotic BFSY slope: for $m = \alpha N$ Boolean quadrics the semi-regular degree of regularity is $D \approx c(\alpha)N$ with

$$c(3) = -3 + \frac{1}{2} + \frac{1}{2} \sqrt{2 \cdot 9 - 30 - 1 + 2 \cdot 5\sqrt{15}} \approx 0.036,$$

i.e. one degree per 28 variables asymptotically, one per 20 in the computed range. The actual thresholds follow the semi-regular ones at 93–97% for $D \geq 4$. Unless the lag grows unexpectedly, the degree of regularity of the Weil-descent system of S_3 therefore grows *linearly* in n — about 6 for $53 \leq n \lesssim 70$, 7 to about 90, 8 to about 110 — and not as a constant, as the first fall degree assumption would require, nor logarithmically. The mechanism is the multiplier 3: the ideal contains three times as many quadrics as there are equations, so the system is as easy as a semi-regular system of $3n$ quadrics (for $m = n$ the slope would be $c(1) \approx 0.09$), but no easier.

For the index calculus this changes nothing qualitatively: the decomposition with two points is exponential in n with a smaller constant than the generic estimate, and the relation collection needs $2^{n/2}$ decompositions regardless.

8 Controls

Random quadrics. As noted in Section 7, $3n$ random dense Boolean quadrics reproduce the semi-regular series exactly (no lag). Random *bilinear* quadrics in the same two blocks of variables collapse earlier (fewer monomials are available to bilinear products); the Weil-descent systems, although bilinear in their quadratic part, behave like dense ones because W_2 is not bilinear.

Subfield subspaces. If $n = 2k$ and $V = \mathbb{F}_{2^k}$ is the subfield — the classical setting of Gaudry and Diem for extension degree 2 — the span of the n equations contains k linear polynomials instead of one, $\dim V_2$ rises from $2n - 1 = 31$ to 99 at $n = 16$, and $D_{\text{solve}} = 3$ for every n we tried (16 to 24). This is not a failure of the model but its confirmation on a degenerate case: for $X_0, X_1 \in \mathbb{F}_{2^k}$ the equation $S_3 = 0$ splits over $\mathbb{F}_{2^k}$ into $Z = pS + a_{6,1}$ and $p^2S^2 + qS + a_{6,0} = 0$ with $S = X_0^2 + X_1^2$, $Z = X_0X_1$ and $x(Q)^2 = px(Q) + q$, the second being Frobenius-linear in S ; the decomposition is a linear system followed by one quadratic equation. HKY’s random subspaces avoid exactly this structure.

Instance independence. All formulas of Sections 5–7 were checked on a second instance at (28, 15) and (28, 16) (level 5) and on 378 instances at $n \leq 24$ (levels 3–4): the dimensions agree to the unit.

9 Open questions

1. *Why $3n$ and $2n - N + 1$.* The dimension of the quadratic part of the ideal and the number of mutants were measured, not derived. Both should follow from the $\mathbb{F}_{2^n}$ -linear structure of S_3 : the equations are the coordinates of a single polynomial over $\mathbb{F}_{2^n}$, and the mutants are combinations $\text{Tr}(\gamma \Lambda S_3)$ with Λ ranging over $\mathbb{F}_{2^n}$ -linear forms in the coordinates, whose cubic part vanishes; the count $2n - N + 1$ has the shape of a dimension count ($2n$ unknowns, N constraints, one degree of freedom) that we could not complete.
2. *The second stage.* A closed form for the rank S_D of the products of variables with the degree falls would give lag_D for every D and turn the $D = 6$ threshold, currently bracketed as $66 \leq N \leq 76$, into a prediction. The data for S_4 (exact) and S_5 (nine points) are in the archive.
3. *Semi-regularity of W_2 .* Is the Macaulay span of W_2 semi-regular in the quotient by ℓ , as the first-stage formula asserts, for all D ? (For the complexity of Weil restriction systems in general see [18, 5].) A proof would settle the first stage; the deviation δ_5 shows that the naive statement needs the corrections listed in Section 7.

4. *Beyond two points.* The tools apply verbatim to the Weil descent of S_4 (three points, cubic equations). Whether the quadric — here cubic — part of the ideal again exceeds the number of equations by a fixed factor is the question that would decide the shape of D_{reg} for the decompositions that matter cryptographically.

10 Reproducibility

All systems are generated from seeds recorded in the supplementary archive (<https://github.com/alekssolov/weil-descent-s3-dreg>); the generator (Python 3, standard library and numpy), the Python closure and its verification by Buchberger–Möller, the C closure program with its Macaulay mode, the Buchberger reference implementation with switchable pair criteria, the emulation of M4GB’s criteria, the M4GB patch (as a script and as a unified diff), every input file run in M4GB and every solver log are provided. Timings: level-5 closures at $N = 32$ take 5–13 h and 3.4–3.7 GB on eight threads; the largest solver runs are (40, 30) — 63 h and 8.5 GB — and the square system $n = 40$ — 6 h and 1.5 GB. Two practical remarks for anyone repeating the solver runs are in Section 3.3: the libtool wrapper in `bin/` is not the executable, and the pair criteria of the 2018 release need the correction of pull request #14 for step degrees to be meaningful.

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